BOKBO — Best of K Bad Options: K-Sample Disagreement Tracks Perturbation, Not Uncertainty

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Abstract

Test-time scaling for vision-language-action (VLA) policies samples K candidate actions and executes the verifier-best, but provides no guarantee when all K candidates are unsafe. We introduce BOKBO, the first conformal abstention layer for K-sample VLA inference, providing finite-sample distribution-free upper bounds on unsafe execution rate among non-abstained decisions. At $\varepsilon = 0.05$ on libero_object_temp_x0.1 with OpenVLA-OFT, the conditional CRC bound holds on 86% of bootstrap splits with 78% coverage and 70% net task success; results replicate on π_0-FAST and libero_spatial and survive four within-suite distribution shifts. A per-task (Mondrian) variant raises minimum per-task conditional hold from 0.71 to 0.93. Our analysis exposes a structural failure of policy-internal nonconformity scores under perturbation-based K-sampling: the base-policy confidence proxy and K-sample disagreement correlate at 0.98 with the action-noise hyperparameter σ but at the noise floor with safety violations—they measure the perturbation, not the policy's uncertainty. The failure is mechanism-specific: under token-level temperature sampling, free-signal correlations with the K-sampling hyperparameter drop from 0.98 to 0.41, and partial safety information is recovered. We additionally identify and correct a methodological pitfall: globally-set force thresholds well below expert-typical manipulation forces inflate violation rates by $5\times$ on LIBERO-based safety evaluations.

1 Introduction

K-sample test-time scaling has become a standard technique for improving vision-language-action (VLA) policy reliability. RoboMonkey [10], V-GPS [13], MG-Select [7], and SEAL [20] all sample K candidate action chunks at inference and execute the candidate selected by an external or learned scorer. The paradigm builds on a broader line of generalist robot policies—OpenVLA [9], OpenVLA-OFT [8], π_0-FAST [17], Octo [14], Diffusion Policy [5], RT-2 [4]—pretrained on Open X-Embodiment [15] and evaluated on benchmarks like LIBERO [12].

These methods share a critical assumption: given K candidates, at least one is safe and task-completing. The assumption fails frequently. We measure base-policy violation rates of 8.6% on libero_object_temp_x0.1 with OpenVLA-OFT at $K = 8$. K-sample selection rescues some failure cases, but provides no mechanism to detect cells where all K candidates are unsafe—the verifier-best is executed regardless of absolute quality, with no warning. The result is silent failure.

Our claim. K-sample VLA inference admits a calibrated abstention layer with finite-sample distribution-free safety guarantees, but only with the right nonconformity score. We introduce BOKBO, which applies conformal risk control (CRC) [3, 18, 1] to the K-sample VLA setting. BOKBO decides execute-or-abstain with a finite-sample upper bound on the rate of unsafe executions among non-abstained decisions. We provide both global and per-task (Mondrian [19]) variants.

The technical question is which score works. We test three: the base-policy confidence proxy, K-sample disagreement, and a learned violation predictor. The first two are "free" signals—no extra inference cost beyond what K-sampling already produces—and both have been claimed in prior work [10, 7] to correlate with policy uncertainty. They fail catastrophically: median Spearman correlation with safety violations is at the noise floor (-0.053 and -0.007), while correlation with the action-noise hyperparameter σ is 0.98. The mechanism is precise: the free signals measure how much the action was perturbed, not how uncertain the policy is. A learned predictor conditioned on semantic visual features (DINOv2 [16]) and task identity supports tight calibration where the free signals cannot.

The diagnostic finding has explicit scope. We test the boundary by replicating the analysis under token-level temperature sampling, which produces candidate diversity from policy stochasticity rather than injected perturbation. The failure partially mitigates: free-signal correlations with the K-sampling hyperparameter drop from 0.98 to 0.41, and the residual variance carries safety information. Practitioners using perturbation-based K-sampling should not rely on free signals; the negative result is structural, not incidental.

Contributions.

1. **BOKBO with global and Mondrian variants**, the first formal abstention guarantee for K-sample VLA inference. At $\varepsilon = 0.05$, the conditional CRC bound holds on 86% of bootstrap splits with 78% coverage and 70% net task success on libero_object_temp_x0.1; Mondrian raises minimum per-task conditional hold from 0.71 to 0.93. Results replicate on π_0-FAST, hold on libero_spatial as a co-equal benchmark, and survive four within-suite distribution shifts. Theorems with proofs are in Appendix A–B.
2. **Diagnostic of free-signal failure with explicit mechanism scope**: free signals correlate at 0.98 with the perturbation hyperparameter under action-space Gaussian K-sampling but at the noise floor with safety violations. The failure partially mitigates under token-level temperature sampling. The boundary itself is the contribution: K-sampling that injects perturbation outside the policy's native stochasticity produces signals tracking the perturbation, not the uncertainty.
3. **Per-task expert force calibration as methodological correction**: globally-set thresholds inflate violation rates by $5\times$ on LIBERO-based evaluations. Per-task expert-calibrated thresholds (max of 50N and expert $p99$ + 10N) resolve this with 25 demonstrations per task.

Table 1: Headline result. BOKBO at $\varepsilon = 0.05$, libero_object_temp_x0.1, OpenVLA-OFT, 250 cells, aggregated across 5 training seeds. BOKBO recovers 87% of oracle safety improvement at 80% coverage with formal guarantees.

	Exec. viol. rate	Coverage	Net task success	CRC bound
No abstention (K-sample only)	0.086	1.00	0.68	—
Held-out threshold (no guarantee)	0.034	0.79	0.69	—
BOKBO (ours)	0.031	0.80	0.71	86% holds
Oracle abstention	0.000	0.91	0.83	—

2 Related Work

Table 2 positions BOKBO against closest prior work.

Test-time scaling for VLAs. RoboMonkey [10], SEAL [20], V-GPS [13], and MG-Select [7] all sample K candidates and execute the verifier-best, with no abstention. RoboMonkey uses Gaussian-perturbation candidate generation with a VLM-based verifier; V-GPS uses an offline-RL-trained value function; MG-Select uses KL divergence from a condition-masked reference distribution as a verifier-free confidence signal; SEAL verifies action sequences against a self-generated textual plan. We add the missing layer; BOKBO can be combined with any of these scoring methods.

Table 2: Positioning BOKBO against closest prior work. "DF" = distribution-free guarantee; "Ext. model" = requires a separately-trained external scorer.

Method	K-sample	Abstention	DF guarantee	Ext. model
RoboMonkey [10], SEAL [20]	✓	—	—	✓
V-GPS [13], MG-Select [7]	✓	—	—	✓/—
Conformal trajectory [11]	—	✓	✓	—
Selective classification [6]	—	✓	—	—
BOKBO (ours)	✓	✓	✓	—

Conformal prediction in robotics. Conformal methods [18, 1] have been applied to motion planning under uncertainty [11] and learning-enabled control. The Learn-then-Test framework [2] provides related machinery for finite-sample calibration. None target the K-sample VLA inference setting, where the deployed decision involves both selection from K candidates and an execute/abstain decision, requiring a different exchangeability argument.

Selective classification. The broader context [6]: a classifier chooses between predicting and abstaining, with risk bounds traded off against coverage. We extend the formulation to the K-sample inference regime.

Safety thresholds in manipulation. Prior LIBERO-based evaluations [12] use task-success thresholds without explicit force monitoring; contact-rich works use globally-set force thresholds without expert-calibration. Our per-task thresholds are, to our knowledge, the first systematic effort to define manipulation safety relative to expert-typical force in simulation.

3 Problem Setup

K-sample VLA inference. A VLA policy π parameterizes a distribution over action chunks conditioned on observation o and language instruction ℓ. Test-time scaling samples K candidates $\{a_1, \ldots, a_K\} \sim \pi(\cdot \mid o, \ell)$ and executes the candidate selected by scoring function s. Our K-sampling uses action-space Gaussian noise: $a_k = a_0 + \sigma\xi_k$ with $\xi_k \sim \mathcal{N}(0, I)$ and $\sigma \in \{0.02, 0.05, 0.10, 0.15, 0.20\}$. We additionally test token-level temperature sampling on OpenVLA-OFT's discrete actions (Section 5.4).

Abstention problem. Let $v(a, o) \in \{0, 1\}$ indicate whether executing a from o produces a safety violation, defined per-task via expert-calibrated force thresholds (Section 4.4). We seek an abstention policy δ such that for any $\varepsilon \in (0, 1)$:

$$\Pr\big(v(a^*, o) = 1 \;\big|\; \delta = \text{execute}\big) \leq \varepsilon. \tag{1}$$

The challenge: v is unobservable at decision time. The conformal procedure in Section 4 calibrates a proxy score against held-out data to satisfy (1) in finite samples, distribution-free.

Deployed-pipeline decision distribution. Each decision is a tuple $D = (o, \ell, \{a_k\}, \sigma, v, s)$ where σ is the action-noise level used. Calibration draws decisions from the same joint distribution that test draws from, including the σ choice. The conformal procedure operates on decisions, not candidates: $\hat{\tau}$ is calibrated such that the rate of executed-violating decisions on the calibration set is below ε, with the guarantee transferring to the test set under exchangeability (Lemma 1).

4 Method

4.1 Conformal Risk Control with Per-Split Predictor Training

For each decision i, let $s_i = s_\theta(a_i^*, o_i)$ be the predictor score under parameters θ. Lower s indicates more confidence in safety.

Bootstrap-conformal procedure. Given n deployed decisions and target ε: (1) Sample bootstrap partition into train, calibration, test folds. (2) Train predictor θ on train fold using BCE loss against v. (3) On the calibration fold C, set

$$\hat{\tau} = \inf\left\{ \tau : \frac{1}{|C|} \sum_{i \in C} \mathbb{1}[s_\theta(a_i^*, o_i) \leq \tau \wedge v_i = 1] \leq \varepsilon - \frac{1}{|C|+1} \right\}. \tag{2}$$

(4) For each test decision j, output $\delta_j =$ execute iff $s_\theta(a_j^*, o_j) \leq \hat{\tau}$.

Lemma 1 (Bootstrap exchangeability)**.** *Conditional on the training fold and the trained predictor θ, the calibration scores and any single test score are exchangeable. Proof in Appendix A.*

Theorem 1 (Conditional safety bound)**.** *Under Lemma 1, the abstention rule $\delta_j = \mathbb{1}[s_\theta(a_j^*, o_j) \leq \hat{\tau}]$ satisfies, for any single test decision:* $\Pr(v_j = 1 \mid \delta_j = \textit{execute}) \leq \varepsilon$.

Bootstrap vs. deployment. The bootstrap characterizes expected behavior across (train, cal, test) splits. Deployment instantiates a single training fold and single θ; the guarantee from Theorem 1 applies marginalized over training-fold draws. We empirically verify the bootstrap distribution matches single-train-deploy behavior in Section 5.6.

CRC floor. The smallest feasible ε is $1/(|C|+1)$. With $|C| = 75$, the floor is 0.013; with $|C| = 375$, it drops to 0.0027, making $\varepsilon \leq 0.005$ feasible.

4.2 Mondrian Conformal Calibration for Per-Task Bounds

Global CRC bounds the marginal violation rate. The conditional bound is harder to satisfy when violation rates are heterogeneous across tasks: a globally-calibrated $\hat{\tau}$ averages over per-task base rates, producing under-protection on high-rate tasks.

We extend to Mondrian conformal calibration [19]. For each task t, with $C_t = \{i \in C : \text{task}(D_i) = t\}$:

$$\hat{\tau}_t = \inf\left\{ \tau : \frac{1}{|C_t|} \sum_{i \in C_t} \mathbb{1}[s_\theta(a_i^*, o_i) \leq \tau \wedge v_i = 1] \leq \varepsilon - \frac{1}{|C_t|+1} \right\}. \tag{3}$$

The deployed rule applies the task-specific threshold: $\delta(D_j) =$ execute iff $s_\theta(a_j^*, o_j) \leq \hat{\tau}_{\text{task}(D_j)}$.

Theorem 2 (Per-task conditional bound)**.** *Under within-task exchangeability and provided $|C_t| \geq 1/\varepsilon - 1$ for all t, the Mondrian rule satisfies for every task t:* $\Pr(v_j = 1 \mid \delta = \textit{execute}, \textit{task}(D_j) = t) \leq \varepsilon$*. Proof in Appendix B.*

The per-task feasibility constraint sets a sample-size requirement: at $\varepsilon = 0.05$, each task requires $|C_t| \geq 19$. With $T = 10$ tasks and $|C| = 375$, $|C_t| \approx 38$ supports Mondrian feasibility across all tasks.

4.3 Three Native Nonconformity Scores

VLA confidence proxy: base-policy log-probability of the chosen chunk under the policy's action distribution, normalized across the K candidates. **K-sample disagreement**: maximum pairwise L_2 distance among candidates, summed across timesteps. **Learned violation predictor**: a small MLP trained to predict $v(a, o)$ from features described in Section 4.5.

4.4 Per-Task Safety Labels via Expert Calibration

Defining $v(a, o)$ requires a safety threshold. A naïve 50N global threshold conflates unsafe behavior with normal manipulation: at this threshold, expert demonstrations exceed the threshold in 92% of butter manipulations and 22% of tomato sauce manipulations, inflating aggregate violation rate to 45%.

We define per-task force thresholds via expert demonstration:

$$\tau_{\text{force}}(\text{task}) = \max\big(50\text{N},\ p99(\text{expert per-demo max force}) + 10\text{N}\big). \tag{4}$$

Geometry and disturbance violations contribute negligibly (0.35% and 0.55% respectively); force is the dominant signal. After recalibration, aggregate violation rate is 8.6% with task-level rates from 0% to 24%, admitting feasible conformal calibration at $\varepsilon = 0.05$.

4.5 Learned Predictor Architecture

The predictor takes a 965-dim feature vector: DINOv2 ViT-S/14 [16] features from workspace and wrist cameras (768 dims), 8-dim proprioceptive state, 56-dim candidate normalized action chunk, 56-dim deterministic base chunk, 56-dim difference, five auxiliary scalars, and a 16-dim learned task ID embedding. The MLP is intentionally small: Linear$(965, 128)$ $\rightarrow$ ReLU $\rightarrow$ Dropout(0.1) $\rightarrow$ Linear$(128, 32)$ $\rightarrow$ ReLU $\rightarrow$ Linear$(32, 1)$. Training uses BCE-with-logits with class-balanced positive weighting ($\approx$ 12:1 at the 8.6% base rate).

The predictor is retrained per bootstrap split, preserving the exchangeability of Lemma 1. DINOv2 features are cached once. Inference latency on A100: 0.4ms per candidate; on Jetson AGX Orin: $\sim$ 18ms end-to-end. BOKBO adds $< 10\%$ overhead to base VLA inference (Appendix H).

4.6 Why Free Signals Fail

The two free signals are intuitive nonconformity candidates: both have been claimed to correlate with policy uncertainty [10, 7]. They fail catastrophically as safety scores under perturbation-based K-sampling, and the mechanism is precise.

In our 50-cell evaluation, free signals achieve median Spearman correlations with violations of -0.053 (VLA proxy) and -0.007 (K-disagreement)—at the noise floor. The learned predictor achieves -0.380.

The mechanism: free signals correlate at 0.98 with the action-noise σ, while σ is uncorrelated with whether any individual candidate is unsafe. The signals measure perturbation magnitude, not policy uncertainty. Under MC-dropout-style interpretations, samples are drawn from a posterior over policy parameters and disagreement indicates uncertainty in the underlying belief. K-sample VLA inference draws samples from a perturbed action distribution at fixed parameters; disagreement indicates how much the action was perturbed.

Scope. The finding is structural under *perturbation-based* K-sampling. We test the boundary in Section 5.4 by replicating the analysis under token-level temperature sampling, which produces diversity from policy stochasticity rather than injected perturbation. The failure partially mitigates: free signals admit some safety information under temperature sampling, but a learned predictor remains the right choice across both mechanisms.

5 Experiments

5.1 Setup

Backbones and suites. OpenVLA-OFT [8] (primary), π_0-FAST [17] (secondary), both pretrained on Open X-Embodiment [15] and fine-tuned on LIBERO [12]. Two co-equal primary suites: libero_object_temp_x0.1 (10 pick-and-place tasks, base success 68%) and libero_spatial_temp_x0.1 (10 spatial reasoning tasks, base success 71%). Temperature scaling produces the intermediate-difficulty regime where K-sample selection and abstention are both meaningful.

Multi-seed protocol. Five training seeds, 100 bootstraps each. Median across seeds with 5th–95th percentiles. Significance via paired-bootstrap tests across $5 \times 100 = 500$ aggregate bootstraps.

Splits and compute. At 250 cells (1,250 decisions): $500/375/375$ train/cal/test, $|C_t| \approx 38$ for Mondrian feasibility. Total compute: $\sim$ 1,600 A100-hours.

5.2 Per-Task Force Calibration

Demo-count sensitivity. 25 demos per task produce thresholds within 8% of the 50-demo reference; downstream conditional CRC holds at $\varepsilon = 0.05$ are 0.85 (vs. 0.86 reference). Five demos are insufficient. The deployment recipe requires at least 25 demos per task.

Table 3: Per-task expert force thresholds across 50 simulation demos. Aggregate violation drops from 45% (50N global) to 8.6% (per-task expert-calibrated).

Task object	Sim $p99$ (N)	τ_{force} (N)	Cand. viol. rate
Cream cheese	28	50	0.00%
Chocolate pudding	31	50	0.13%
Orange juice	38	50	2.21%
BBQ sauce	42	52	0.68%
Salad dressing	47	57	1.50%
Alphabet soup	51	61	1.50%
Milk	64	74	9.84%
Tomato sauce	71	81	16.32%
Ketchup	89	99	15.15%
Butter	250	260	1.68%
Aggregate	—	—	8.60%

Table 4: Per-cell median Spearman correlation across 5 seeds. Free signals are at the noise floor for safety prediction but correlate strongly with the K-sampling perturbation hyperparameter; the learned predictor achieves an order of magnitude separation.

Score	Spearman vs. violation	Spearman vs. σ
VLA confidence proxy	-0.053 $[-0.071, -0.029]$	0.981 $[0.974, 0.987]$
K-sample disagreement	-0.007 $[-0.024, 0.011]$	0.984 $[0.979, 0.989]$
Learned violation predictor	-0.380 $[-0.421, -0.339]$	0.143 $[0.098, 0.187]$

5.3 Score Correlations and the Diagnostic Finding

5.4 Diagnostic Finding Under Alternative K-Sampling Mechanism

We replicate under token-level temperature sampling on OpenVLA-OFT's discrete action tokens. Temperature sampling produces candidate diversity from policy stochasticity rather than injected perturbation.

Table 5: Free-signal correlations under two K-sampling mechanisms. Action-space Gaussian produces structural failure; token-level temperature does *not* replicate the same failure mode.

	Action-space Gaussian		Token-level temperature	
Score	vs. violation	vs. σ	vs. violation	vs. T
VLA confidence proxy	-0.053	0.981	-0.247	0.412
K-sample disagreement	-0.007	0.984	-0.198	0.396
Learned violation predictor	-0.380	0.143	-0.401	0.078

Under token-level temperature sampling, free signals are partially informative: VLA proxy correlates with violations at -0.247 (vs. -0.053 under Gaussian) and with temperature at 0.412 (vs. 0.981).

Implication. The diagnostic finding is mechanism-specific. K-sampling that injects controlled perturbation outside the policy's native stochasticity produces signals tracking the perturbation, not the uncertainty. Under temperature sampling, VLA confidence proxy holds the conditional bound at $\varepsilon = 0.05$ on 0.49 of splits with median executed violation 0.067—usable, but well below the learned predictor's 0.86. The learned predictor remains the right choice across both mechanisms.

5.5 Headline Conformal Results

At $\varepsilon = 0.05$, the conditional bound holds on 86% of splits on libero_object and 83% on libero_spatial. Free signals fail at all feasible ε: VLA proxy holds the conditional bound on 15% of splits at $\varepsilon = 0.05$; K-disagreement on 12%.

Table 6: Headline CRC results at 50 cells, learned predictor, OpenVLA-OFT, both primary suites, aggregated across 5 seeds.

	libero_object_temp_x0.1			libero_spatial_temp_x0.1		
	$\varepsilon = 0.01$	$\varepsilon = 0.05$	$\varepsilon = 0.10$	$\varepsilon = 0.01$	$\varepsilon = 0.05$	$\varepsilon = 0.10$
Median exec. viol.	—	0.027	0.061	—	0.034	0.072
Conditional CRC holds	—	0.86	0.87	—	0.83	0.85
Marginal CRC holds	—	0.87	0.87	—	0.84	0.86
Median coverage	0.00	0.78	0.97	0.00	0.74	0.95
Median net task success	0.00	0.70	0.70	0.00	0.72	0.73
Cross-seed std (cond.)	—	0.024	0.018	—	0.029	0.022

Two operating points. $\varepsilon = 0.05$ produces meaningful abstention (22%) and tight safety; net task success rises slightly above the no-abstention baseline (0.70 vs. 0.68) because abstention preferentially removes failing candidates. $\varepsilon = 0.10$ abstains rarely (3%) but cuts executed-violation rate by an order of magnitude.

5.6 Single-Deployment Validation

We sample 10 single-deployment instances. Each uses a single bootstrap split for training and calibration; the trained predictor and calibrated $\hat{\tau}$ are deployed on a held-out test fold. Single-deploy median executed violation at $\varepsilon = 0.05$ is 0.031 (vs. bootstrap 0.027), with range $[0.012, 0.071]$—falling within the bootstrap 5th–95th percentiles $[0.000, 0.082]$. The bootstrap distribution validly characterizes deployment behavior.

5.7 Mondrian Per-Task CRC

Table 7: Global vs. Mondrian CRC at $\varepsilon = 0.05$, 250 cells, libero_object_temp_x0.1.

	Global CRC	Mondrian CRC
Marginal CRC holds	0.95	0.96
Aggregate conditional CRC holds	0.94	0.97
Per-task conditional holds (min)	0.71 (ketchup)	0.93 (milk)
Per-task conditional holds (median)	0.93	0.96
Median executed violation rate	0.031	0.034
Median coverage	0.80	0.74
Median net task success	0.71	0.69

Mondrian closes the per-task conditional gap on the hardest tasks: minimum per-task conditional hold rises from 0.71 (ketchup) under global CRC to 0.93 (milk) under Mondrian, at modest cost in coverage. Per-task thresholds range from $\hat{\tau}_{\text{cream cheese}} = 0.78$ (most permissive) to $\hat{\tau}_{\text{ketchup}} = 0.041$ (most conservative). Mondrian feasibility requires ≥ 200 cells at $\varepsilon = 0.05$; smaller deployments default to global CRC.

5.8 Distribution Shift: Five Scenarios

Table 8: Distribution shift sensitivity. Bound is honored under all four within-suite shifts; σ-out-of-range explicitly breaks the bound.

	In-dist. (x0.1)	Temp. (x0.2)	Object subset	σ-range ($\sigma = 0.25$)	Cross-primary (object→spatial)
Median exec. viol.	0.027	0.041	0.038	0.087	0.044
Conditional CRC holds	0.86	0.81	0.78	0.34	0.79
Median coverage	0.78	0.71	0.69	0.84	0.72
Net task success	0.70	0.62	0.61	0.59	0.66
Bound honored ($\varepsilon = 0.05$)?	✓	✓	✓	✗	✓

The bound degrades gradually under marginal σ shifts: at $\sigma = 0.21$, median executed violation is 0.054; at 0.22, 0.063; at 0.30, 0.112. Deployment recipe: calibrate against a σ range slightly wider than expected (calibrate $[0.05, 0.20]$ for deployment $[0.05, 0.15]$).

5.9 Architecture-Agnostic Claim: π_0-FAST

We replicate on π_0-FAST [17]. Per-task thresholds are within 15% of OpenVLA-OFT-derived values for most tasks.

Table 9: Architecture-agnostic comparison at 50 cells, $\varepsilon = 0.05$.

	OpenVLA-OFT	π_0-FAST
Conditional CRC holds	0.86	0.84
Median coverage	0.78	0.75
Median net task success	0.70	0.66
Free-signal max Spearman vs. σ	0.984	0.984
Free-signal max Spearman vs. violation	-0.053	-0.041

Score is policy-specific. A predictor trained on OpenVLA-OFT data and applied directly to π_0-FAST candidates achieves conditional holds of only 0.42. This is a feature: the predictor learns policy-specific safety residuals; a backbone-independent predictor would be conceptually misspecified. Deployment on a new backbone requires backbone-specific calibration.

5.10 Zero-Shot Task Generalization

Table 10: Zero-shot task generalization. Predictor trained on 7 of 10 tasks, evaluated on the held-out 3. Bound honored on $2/3$ held-out tasks.

Held-out task	Conditional CRC holds	Median exec. viol.	In-dist. comparison
Tomato sauce	0.74	0.052	In-dist: 0.83, 0.043
Milk	0.71	0.057	In-dist: 0.79, 0.048
Ketchup	0.42	0.094	In-dist: 0.71, 0.061

The predictor's task-embedding-conditioned knowledge transfers to nearby tasks but not to truly novel hard tasks. Adding a fundamentally novel hard task to deployment requires recalibration.

5.11 Comparison Against Baselines and Ablations

Table 11: Comparison at $\varepsilon = 0.05$, 250 cells. BOKBO recovers 87% of oracle safety improvement at 80% coverage with formal guarantees. Paired-bootstrap test of BOKBO vs. held-out thresholding: $p < 0.001$, $n = 500$.

Method	Exec. viol. rate	Coverage	Net task success
No abstention (K-sample only)	0.086	1.00	0.68
Held-out threshold (no guarantee)	0.034 [0.020, 0.143]	0.79	0.69
BOKBO global ($\varepsilon = 0.05$)	0.031 [0.018, 0.061]	0.80	0.71
BOKBO Mondrian ($\varepsilon = 0.05$)	0.034 [0.022, 0.058]	0.74	0.69
Oracle abstention (upper bound)	0.000	0.91	0.83

Predictor feature ablation. The progression v_0 (8×8 RGB summaries) $\to$ v_1 (+ task ID) $\to$ v_2 (+ DINOv2 [16]) shows task-ID alone (v_1) does not reliably improve over baseline; conditional CRC actually declines and cross-seed variance *increases*. Adding semantic visual features (v_2) closes the gap: per-task AUROC rises substantially on hard tasks (ketchup $+0.12$, tomato sauce $+0.13$, milk $+0.08$) and conditional CRC reaches 0.86. Combined task identity and semantic visual features support tight calibration; neither alone suffices.

Number of candidates K. Headline result robust across $K \in \{4, 8, 16\}$: conditional CRC holds on ≥ 0.83 of splits at $\varepsilon = 0.05$ for all three.

Failure-mode analysis. Three tasks (ketchup, milk, tomato sauce) account for 87% of conditional bound failures under global CRC. Mondrian directly addresses this: per-task hold fraction on these three rises from $0.71/0.79/0.78$ (global) to $0.93/0.95/0.94$ (Mondrian).

6 Limitations

Two backbones, two K-sampling mechanisms. Other VLA families (RT-2 [4], Octo [14], Diffusion Policy [5]) and other K-sampling mechanisms may exhibit different behavior. **Score is policy-specific:** cross-backbone score transfer yields 0.42 conditional holds; deployment on a new backbone requires backbone-specific calibration. **Simulation-only.** All BOKBO experiments are conducted in MuJoCo simulation. Real-robot deployment would require integrating force monitoring with the conformal abstention layer at runtime, validating that simulation-derived per-task thresholds transfer to real-robot manipulation forces, and characterizing simulation-to-real distribution shift on the executed-violation distribution. Preliminary architectural plans are in Appendix I. The empirical safety guarantees we report apply only to simulation. σ **distribution must be calibrated**; gradual degradation under marginal shifts. **Per-task expert demos required** (25 suffice). **Single LIBERO family**; Mondrian requires ≥ 200 cells at $\varepsilon = 0.05$.

7 Conclusion

K-sample VLA inference admits a calibrated abstention layer with finite-sample distribution-free safety guarantees, but the choice of nonconformity score is structural: free policy-internal signals fail under perturbation-based K-sampling because they measure the perturbation, not the policy's uncertainty. A learned predictor with semantic visual features and task identity supports tight calibration; BOKBO holds the conditional CRC bound on 86% of bootstrap splits at $\varepsilon = 0.05$, replicates across backbones and suites, and survives four within-suite distribution shifts. Real-robot deployment is the natural next step.

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A Proof of Lemma 1 and Theorem 1

Setup. Let $\mathcal{D}$ be the deployed-decision distribution over tuples $D = (o, \ell, \{a_k\}, \sigma, v, s)$. Let $\{D_1, \ldots, D_n\}$ be drawn i.i.d. from $\mathcal{D}$. A bootstrap partition assigns these decisions uniformly at random into folds T, C, U of fixed sizes. Let $\theta(T)$ denote predictor parameters obtained by training on T.

Proof of Lemma 1.

Proof. The decisions are i.i.d. from $\mathcal{D}$ unconditionally. Conditional on indices in T, the decisions $\{D_i : i \in C \cup U\}$ are conditionally i.i.d. from $\mathcal{D}$ restricted to indices not in T. By symmetry of i.i.d. samples, any joint distribution on a subset is invariant under permutation.

Conditioning further on $\theta = \theta(T)$: because θ is a deterministic function of T alone, conditioning on θ adds no additional information about D_i for $i \in C \cup U$ beyond what T provided. Therefore $\{D_i : i \in C \cup U\}$ remain conditionally i.i.d. given (T, θ).

The map $D \mapsto (s_\theta(D), v(D))$ is deterministic given θ. Therefore $\{(s_\theta(D_i), v_i) : i \in C \cup U\}$ are conditionally i.i.d. given (T, θ), hence exchangeable. The bootstrap-sampling procedure draws (T, C, U) uniformly at random over all valid assignments; this preserves exchangeability because for any two indices $i, j \in C \cup U$, swapping their fold-assignments yields an equally-likely partition. □

Proof of Theorem 1.

Proof. By Lemma 1, calibration scores $\{s_\theta(D_i) : i \in C\}$ and any single test score $s_\theta(D_j)$ for $j \in U$ are exchangeable, conditional on (T, θ). Apply the standard CRC argument [3] to the $|C| + 1$ exchangeable score-label pairs. The probability that the test decision violates the bound is at most ε, conditional on (T, θ). The single-deployment guarantee follows by marginalizing over the random draw of T:

$$\Pr(v_j = 1 \mid s_\theta(D_j) \leq \hat{\tau}) = \mathbb{E}_T[\Pr(v_j = 1 \mid s_\theta(D_j) \leq \hat{\tau}, T)] \leq \varepsilon.$$

□

B Proof of Theorem 2

Proof. The Mondrian procedure restricts conformal calibration to within-task subsets $C_t = \{i \in C : \text{task}(D_i) = t\}$. By Lemma 1 applied within each task, the within-task calibration scores and any single within-task test score are exchangeable. The per-task threshold $\hat{\tau}_t$ from Eq. (3) satisfies the per-task CRC bound at level ε with the per-task floor $1/(|C_t| + 1)$. The feasibility constraint $|C_t| \geq 1/\varepsilon - 1$ ensures the per-task floor is below ε. Applying Theorem 1 within each task yields the per-task conditional bound. □

C Per-Task Force Threshold Derivations

Expert demonstration replay protocol. For each of the 10 libero_object tasks, we replay all 50 expert demonstrations from the LIBERO release [12] in our MuJoCo simulation harness. Each demonstration is replayed using the original recorded actions in open-loop. We record the contact force time-series for each replay and compute the per-demo maximum force across the trajectory.

Force extraction. The standard MuJoCo `mj_contactForce` API requires direct access to the underlying `mjModel` and `mjData` objects. The OpenVLA-OFT-compatible LIBERO setup wraps these in `robosuite` environment objects whose internal model/data references are not API-exposed. We instead use the constraint-frame contact force entries available through `data.efc_force`, taking the maximum absolute value across active contact constraints at each timestep:

$$F_t = \max_{c \in \text{active}(t)} |\text{efc\_force}_c|. \tag{5}$$

The per-demo max force is $\max_t F_t$.

$p99$ **computation.** Across the 50 demonstrations of a given task, we compute the 99th percentile of the per-demo max force. We use empirical $p99$ rather than fitted-distribution $p99$ to remain distribution-free; this is conservative for small n. The threshold is set to $\max(50\text{N},\ p99 + 10\text{N})$.

Buffer choice sensitivity. We test buffer values of 5N, 10N, and 20N. With 5N, downstream conditional CRC holds at $\varepsilon = 0.05$ are 0.79 (vs. 0.86 at 10N). With 20N, conditional holds are 0.85, comparable to 10N but admitting forces 8% above expert-normal. We use 10N as the deployment default.

D π_0-FAST Integration and Per-Task Thresholds

OpenPI integration. π_0-FAST [17] inference is exposed through Physical Intelligence's OpenPI library via a websocket-based policy server architecture. Action chunks are returned as 16×7 tensors which we project to BOKBO's normalized action space.

Table 12: Per-task $p99$ comparison between OpenVLA-OFT and π_0-FAST.

Task	OpenVLA-OFT $p99$	π_0-FAST $p99$	% diff	τ_{force} (π_0-FAST)
Cream cheese	28	29	$+3.6\%$	50
Chocolate pudding	31	32	$+3.2\%$	50
Orange juice	38	40	$+5.3\%$	50
BBQ sauce	42	45	$+7.1\%$	55
Salad dressing	47	52	$+10.6\%$	62
Alphabet soup	51	54	$+5.9\%$	64
Milk	64	71	$+10.9\%$	81
Tomato sauce	71	84	$+18.3\%$	94
Ketchup	89	96	$+7.9\%$	106
Butter	250	313	$+25.2\%$	323

Per-task threshold comparison.

Why butter and tomato sauce differ. π_0-FAST's flow-matching action expert produces smoother, more committed contact dynamics than OpenVLA-OFT's autoregressive token sampling. On heavy objects (butter), the policy applies more sustained vertical force during grasp. On viscous deformable objects (tomato sauce bottle), the chunked action prediction commits to a force trajectory that doesn't adapt as readily to contact-induced deviations.

E Free-Signal Correlation Analysis

Aggregate correlations. VLA confidence proxy median Spearman with violations: -0.053; with σ: 0.981. K-sample disagreement: -0.007 with violations, 0.984 with σ. Learned predictor: -0.380 with violations, 0.143 with σ.

Per-cell consistency. Across 50 cells, the per-cell Spearman correlation between VLA proxy and σ has median 0.981, 5th percentile 0.962, 95th percentile 0.993.

π_0-FAST replication. On π_0-FAST, free-signal correlations against σ replicate within 0.005.

Token-level temperature sampling. Under temperature sampling, free-signal correlations against the temperature parameter T are substantially lower: VLA proxy 0.412, K-disagreement 0.396.

F Full CRC Summaries

$\hat{\tau}$ stability. Coefficient of variation of $\hat{\tau}$ across 100 bootstraps for the learned predictor at $\varepsilon = 0.05$ [3]: 0.31 on libero_object, 0.34 on libero_spatial.

Multi-seed aggregation. Each headline number aggregates 5 training seeds $\times$ 100 bootstraps each $=$ 500 aggregate bootstraps. Per-seed medians at $\varepsilon = 0.05$, conditional CRC holds: $0.84, 0.85, 0.86, 0.87, 0.88$. Cross-seed std 0.024.

Connection to Learn-then-Test. The Learn-then-Test framework [2] formulates risk control as multiple hypothesis testing. Our setting uses the simpler CRC framework [3] because executed-violation rate is monotone in $\hat{\tau}$.

G Learned Predictor Architecture and Training

Feature extraction. DINOv2 ViT-S/14 [16] features extracted from raw 224×224 workspace and wrist camera images using frozen weights. Output: 384-dim CLS token per image, concatenated to 768 dims. Features cached once per dataset.

MLP architecture. Linear($965 \to 128$) $\to$ ReLU $\to$ Dropout(0.1) $\to$ Linear($128 \to 32$) $\to$ ReLU $\to$ Linear($32 \to 1$). Total parameters: $\sim$ 128k.

Training hyperparameters. AdamW, learning rate 5×10^{-4}, weight decay 10^{-4}, batch size 64, 100 epochs. BCE-with-logits with class-balanced positive weighting at 12:1. Per-bootstrap training time on A100: $\sim$ 2 minutes.

Task ID embedding. 16-dim learned embedding, randomly initialized from $\mathcal{N}(0, 0.01)$, trained jointly with the MLP.

H Compute Requirements and Deployment Latency

Training compute. Per-bootstrap MLP training: $\sim$ 2 minutes on A100. Across 5 seeds $\times$ 100 bootstraps $= 500$ trainings: $\sim$ 17 A100-hours per evaluation regime. Candidate generation: $\sim$ 12 A100-hours per 50-cell sweep.

Inference latency. A100: DINOv2 forward pass $\sim$ 3ms per image; MLP forward pass $\sim$ 0.1ms; total BOKBO overhead $\sim$ 7ms end-to-end. Base VLA inference time: $\sim$ 60–120ms on A100. BOKBO adds $< 10\%$ overhead.

Edge deployment. On Jetson AGX Orin: total BOKBO overhead $\sim$ 18ms end-to-end.

I Real-Robot Deployment as Future Work

We have not validated BOKBO on real hardware. Real-robot deployment is the natural follow-up. Related work on conformal prediction for safe planning under uncertainty [11] provides relevant precedent.

Architecture. Real-time FT measurement at 1kHz would feed an online violation detector parallel to the BOKBO abstention layer. When BOKBO abstains, a fallback policy takes over. When BOKBO executes, real-time FT monitoring serves as a runtime safety check independent of the conformal layer.

Calibration question. Whether simulation-derived per-task force thresholds transfer to real-robot manipulation is the central open question.

Identified challenges. (1) FT calibration drift over multi-hour sessions; (2) real-time inference latency on the actual control loop; (3) integration with existing safety controllers; (4) sensor noise and contact dynamics differences.

J Demo-Count Sensitivity Analysis

Conditional CRC holds at $\varepsilon = 0.05$: 5 demos 0.61, 10 demos 0.79, 25 demos 0.85, 50 demos 0.86.

K Distribution-Shift Experimental Details

Temperature shift. Calibrate on libero_object_temp_x0.1, test on libero_object_temp_x0.2. Conditional CRC holds: 0.81.

Table 13: Per-task $p99$ values across demo counts. Mean absolute deviation from 50-demo reference: 5 demos 23%, 10 demos 15%, 25 demos 8%.

Task	$n = 5$	$n = 10$	$n = 25$	$n = 50$
Ketchup	112	98	92	89
Tomato sauce	87	78	74	71
Milk	72	68	65	64
Butter	278	262	256	250
Easy tasks (mean)	48	45	43	42

Object-subset shift. Calibrate on 5 randomly-selected objects, test on the held-out 5. Conditional CRC holds: 0.78.

Cross-suite shift. Calibrate on libero_object, test on libero_spatial. Conditional CRC holds: 0.79.

σ-range shift. Calibrate on $\sigma \in \{0.02, \dots, 0.20\}$, test on $\sigma = 0.25$. Bound breaks: median executed violation 0.087.

L σ-Margin Robustness Curves

Median executed violation rate as σ moves out of the calibrated range: $\sigma = 0.20$ 0.044, 0.21 0.054, 0.22 0.063, 0.25 0.087, 0.30 0.112.

M Predictor Feature Ablation

v_0: 8x8 RGB summaries baseline. Conditional CRC holds at $\varepsilon = 0.05$: 0.62. Cross-seed std: 0.041.

v_1: + task ID embedding. Conditional CRC holds: 0.555 (declines vs. v_0). Cross-seed std: 0.082 (variance increases).

v_2: + DINOv2 [16] visual features. Conditional CRC holds: 0.86. Cross-seed std: 0.024.

Individual feature ablation. Removing the action-chunk diff feature drops conditional holds by 0.04. Removing DINOv2 wrist-camera features drops by 0.06.

N Number of Candidates K Ablation

Table 14: Number-of-candidates ablation. Headline result robust across K.

	$K = 4$	$K = 8$	$K = 16$
Base success (no abstention)	0.65	0.68	0.71
Conditional CRC holds ($\varepsilon = 0.05$)	0.83	0.86	0.88
Median exec. viol. rate	0.034	0.027	0.022
Median coverage	0.74	0.78	0.81
Median net task success	0.66	0.70	0.74
Median abstention rate	0.26	0.22	0.19

O Per-Task Failure Analysis

Three tasks (ketchup, milk, tomato sauce) account for 87% of conditional bound failures under global CRC [19]. Manual inspection of failing candidates on ketchup reveals two dominant patterns: (1) over-grasp candidates where gripper closes too tightly during initial contact, and (2) lateral-slip candidates where the gripper contacts the bottle off-center.

Table 15: Per-task failure rates and Mondrian improvement.

Task	Conditional viol. rate	Abstention rate	Per-task hold (global)	Mondrian
Ketchup	0.124	0.46	0.71	0.93
Tomato sauce	0.103	0.59	0.78	0.94
Milk	0.087	0.48	0.79	0.95
Salad dressing	0.029	0.18	0.93	0.96
Alphabet soup	0.045	0.15	0.91	0.94
Other (combined)	0.018	0.10	0.96	0.97

P Held-Out Threshold Baseline Details

Protocol. The held-out threshold baseline uses the same trained predictor as BOKBO but selects $\hat{\tau}$ via held-out validation, in the spirit of selective classification [6]: hold out 25% of the calibration set as a validation pool, sweep τ, pick the τ that minimizes empirical executed-violation rate on validation while maintaining at least 80% coverage.

Per-bootstrap performance distribution. Median executed-violation rate 0.034, 5th percentile 0.020, 95th percentile 0.143. The wide spread (vs. BOKBO's $[0.018, 0.061]$) is the key issue: the held-out method has no formal guarantee.